MPG No. 11.5.39 $4'/m'$ [Type III, tetragonal] [G tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & G_z \end{bmatrix}$$

$$G_z = \mathbb{G}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & G_{xxz} \\ 0 & 0 & G_{xxz} \\ 0 & 0 & G_{zzz} \\ 0 & G_{xxz} & 0 \\ G_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxz} = \mathbb{G}_z^{(1)} - \mathbb{G}_z^{\alpha(1)}$$
$$G_{zzz} = 3\mathbb{G}_z^{(1)} + 2\mathbb{G}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & G_{xxz} \\ 0 & 0 & G_{xxz} \\ 0 & 0 & 0 \\ G_{yzx} & -\frac{G_{xxz}}{2} & 0 \\ -\frac{G_{xxz}}{2} & -G_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxz} = 2\mathbb{G}_z^{(2)}$$
$$G_{yzx} = 3\mathbb{Q}_u^{(1)}$$
$$\frac{G_{xxz}}{2} = \mathbb{G}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & G_{yzy} & 0 \\ -G_{yzy} & G_{yzx} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)}$$
$$G_{yzy} = \mathbb{G}_z^{(3)}$$
$$G_{xyz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).